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Chapter 17 - Reduction To First-Order Systems

Earlier chapters treated higher-order differential equations and systems as related subjects. This chapter makes their connection explicit: every linear differential equation of order n can be rewritten as a system of n first-order equations.

The conversion does not approximate the original problem. It repackages the same unknown function and its derivatives into a vector called the **state**.

The main questions are:

- why does an n th-order equation need n state variables?
- how should those variables be chosen?
- how is the highest derivative isolated without changing signs incorrectly?
- where do the coefficient matrix and forcing vector come from?
- how are initial conditions transferred into an initial state vector?
- how are coupled equations handled when their orders differ?
- how can the matrix system be checked against the original equations?
- when can Chapter 16's matrix exponential solve the converted system?

From Derivative Memory To A State Vector

The Goal Of This Block

The goal is to understand why a higher-order equation can be represented by several first-order equations without losing information.

What A Higher-Order Equation Must Remember

Consider a second-order equation for an unknown function $y(t)$. To predict how the solution changes, it is generally not enough to know only $y(t)$. We also need its current rate $y'(t)$.

For a third-order equation, the required current information is:

$$y(t), \quad y'(t), \quad y''(t).$$

For an n th-order equation, the corresponding list is:

$$y(t), \quad y'(t), \quad y''(t), \quad \dots, \quad y^{(n-1)}(t).$$

The key intuition:

an n th-order scalar equation stores its present state in n quantities: the function and its first $n - 1$ derivatives.

Building The State Vector

For a third-order equation, define:

$$x_1(t) = y(t), \quad x_2(t) = y'(t), \quad x_3(t) = y''(t).$$

Collect the three derivative levels into one column vector:

$$\mathbf{x}(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} = \begin{bmatrix} y(t) \\ y'(t) \\ y''(t) \end{bmatrix}.$$

This vector is the **state vector**. Bold $\mathbf{x}$ denotes the whole vector; x_1, x_2, x_3 denote its scalar components.

The Derivative Chain

Differentiate each state definition.

From $x_1 = y$:

$$x'_1 = y' = x_2.$$

From $x_2 = y'$:

$$x'_2 = y'' = x_3.$$

From $x_3 = y''$:

$$x'_3 = y'''.$$

The first two equations merely pass information along the derivative chain. The original differential equation is used only when replacing the final quantity y''' .

A useful interpretation:

most rows of the matrix system record definitions; the final row contains the actual dynamics supplied by the original equation.

Why The Conversion Is Equivalent

Suppose $y(t)$ solves the original third-order equation. Defining:

$$x_1 = y, \quad x_2 = y', \quad x_3 = y''$$

automatically makes $x'_1 = x_2$ and $x'_2 = x_3$. The original equation then determines $x'_3 = y'''$. Therefore y produces a solution of the first-order system.

Conversely, suppose a system solution satisfies:

$$x'_1 = x_2, \quad x'_2 = x_3.$$

Set $y = x_1$. The first relation gives $y' = x_2$. Differentiate that relation:

$$y'' = x'_2 = x_3.$$

The final system equation then becomes the original equation for y''' . Thus the two descriptions contain the same solution information.

Block 1 Summary

For an n th-order equation, use n state variables:

$$x_1 = y, \quad x_2 = y', \quad \dots, \quad x_n = y^{(n-1)}.$$

For this n -component state, the first $n - 1$ derivatives form the chain:

$$x'_1 = x_2, \quad x'_2 = x_3, \quad \dots, \quad x'_{n-1} = x_n.$$

The original equation supplies the final equation for $x'_n = y^{(n)}$.

A Complete Second-Order Conversion

The Goal Of This Block

The goal is to convert one second-order initial-value problem slowly enough that every matrix entry can be traced back to a visible scalar equation.

The Problem

Convert:

$$2y'' - 3y' + 8y = 4 \cos t,$$

with:

$$y(0) = 1, \quad y'(0) = -2,$$

into:

$$\mathbf{x}' = A\mathbf{x} + \mathbf{f}(t), \quad \mathbf{x}(0) = \mathbf{c}.$$

Here A is the coefficient matrix, $\mathbf{f}(t)$ is the forcing vector, and $\mathbf{c}$ is the initial state.

Step 1: Isolate The Highest Derivative

Start from:

$$2y'' - 3y' + 8y = 4 \cos t.$$

Move the y' and y terms to the right side:

$$2y'' = 3y' - 8y + 4 \cos t.$$

Divide every term by 2:

$$y'' = -4y + \frac{3}{2}y' + 2 \cos t.$$

Writing $-4y$ before $\frac{3}{2}y'$ anticipates the state order y, y' .

Step 2: Define The State Variables

The equation is second order, so define:

$$x_1 = y, \quad x_2 = y'.$$

The state vector is:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} y \\ y' \end{bmatrix}.$$

Step 3: Differentiate The State Definitions

Differentiate $x_1 = y$:

$$x'_1 = y'.$$

Because $x_2 = y'$:

$$\boxed{x'_1 = x_2}.$$

Differentiate $x_2 = y'$:

$$x'_2 = y''.$$

Use the isolated formula:

$$y'' = -4y + \frac{3}{2}y' + 2 \cos t.$$

Replace y by x_1 and y' by x_2 :

$$\boxed{x'_2 = -4x_1 + \frac{3}{2}x_2 + 2 \cos t}.$$

Step 4: Stack The Scalar Equations

The two first-order equations are:

$$\begin{aligned}x'_1 &= x_2, \\x'_2 &= -4x_1 + \frac{3}{2}x_2 + 2 \cos t.\end{aligned}$$

Separate state-dependent terms from the forcing:

$$\begin{bmatrix} x_1' \\ x_2' \end{bmatrix} = \begin{bmatrix} 0x_1 + 1x_2 \\ -4x_1 + \frac{3}{2}x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 2\cos t \end{bmatrix}.$$

The coefficient of x_1 forms column 1, and the coefficient of x_2 forms column 2. Therefore:

$$\mathbf{x}' = \begin{bmatrix} 0 & 1 \\ -4 & \frac{3}{2} \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 2\cos t \end{bmatrix}.$$

Step 5: Convert The Initial Conditions

The state ordering is:

$$\mathbf{x}(t) = \begin{bmatrix} y(t) \\ y'(t) \end{bmatrix}.$$

Evaluate at $t = 0$:

$$\mathbf{x}(0) = \begin{bmatrix} y(0) \\ y'(0) \end{bmatrix}.$$

Substitute $y(0) = 1$ and $y'(0) = -2$:

$$\mathbf{x}(0) = \begin{bmatrix} 1 \\ -2 \end{bmatrix}.$$

The initial values are not coefficients in A ; they form a separate vector.

Verify The Conversion Row By Row

Start from the proposed matrix system:

$$\mathbf{x}' = \begin{bmatrix} 0 & 1 \\ -4 & \frac{3}{2} \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 2 \cos t \end{bmatrix}.$$

Replace $\mathbf{x}$ by $[x_1, x_2]^T$ and multiply row by row:

$$\begin{aligned} x_1' &= 0x_1 + 1x_2 + 0 = x_2, \\ x_2' &= -4x_1 + \frac{3}{2}x_2 + 2 \cos t. \end{aligned}$$

Restore $x_1 = y$, $x_2 = y'$, and $x_2' = y''$ in the second row:

$$y'' = -4y + \frac{3}{2}y' + 2 \cos t.$$

Multiply every term by 2:

$$2y'' = -8y + 3y' + 4 \cos t.$$

Move the y and y' terms to the left:

$$2y'' - 3y' + 8y = 4 \cos t.$$

This is the original differential equation, so the conversion is correct.

Block 2 Summary

For a second-order equation:

1. isolate y''
2. define $x_1 = y$ and $x_2 = y'$
3. use $x'_1 = x_2$
4. replace y and y' in the isolated formula to obtain x'_2
5. read the matrix coefficients from the two scalar equations
6. place $y(t_0)$ and $y'(t_0)$ into the initial vector in state order

The matrix is assembled from coefficients, not guessed from the original equation.

The General Nth-Order Companion Form

The Goal Of This Block

The goal is to extend the second-order construction to a linear equation of any order and understand the pattern in its coefficient matrix.

Normalize The Differential Equation First

Consider:

$$b_n(t)y^{(n)} + b_{n-1}(t)y^{(n-1)} + \cdots + b_1(t)y' + b_0(t)y = g(t),$$

on an interval where:

$$b_n(t) \neq 0.$$

Divide every term by $b_n(t)$. Define:

$$p_j(t) = \frac{b_j(t)}{b_n(t)}, \quad q(t) = \frac{g(t)}{b_n(t)}.$$

The normalized equation is:

$$y^{(n)} + p_{n-1}(t)y^{(n-1)} + \cdots + p_1(t)y' + p_0(t)y = q(t).$$

Isolate the highest derivative:

$$\boxed{y^{(n)} = -p_0(t)y - p_1(t)y' - \cdots - p_{n-1}(t)y^{(n-1)} + q(t)}.$$

Every coefficient moved from the left side changes sign.

Define The Derivative-Chain State

Use:

$$x_1 = y, \quad x_2 = y', \quad x_3 = y'', \quad \dots, \quad x_n = y^{(n-1)}.$$

Differentiating the first $n - 1$ definitions gives:

$$\begin{aligned} x'_1 &= x_2, \\ x'_2 &= x_3, \\ &\vdots \\ x'_{n-1} &= x_n. \end{aligned}$$

For the final component:

$$x'_n = y^{(n)}.$$

Substitute the isolated highest derivative and replace each derivative by its state component:

$$\boxed{x'_n = -p_0(t)x_1 - p_1(t)x_2 - \dots - p_{n-1}(t)x_n + q(t)}.$$

The Companion Matrix

Collect the state components into:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix}.$$

The system is:

$$\boxed{\mathbf{x}' = A(t)\mathbf{x} + \mathbf{f}(t)},$$

where:

$$A(t) = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -p_0(t) & -p_1(t) & -p_2(t) & \cdots & -p_{n-1}(t) \end{bmatrix}$$

and:

$$\mathbf{f}(t) = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ q(t) \end{bmatrix}.$$

This arrangement is called a **companion form**. The 1s above the main diagonal create the derivative chain. The final row comes from the isolated formula for $y^{(n)}$.

Why The Forcing Appears Only In The Last Row

The equations:

$$x'_1 = x_2, \quad x'_2 = x_3, \quad \dots, \quad x'_{n-1} = x_n$$

are definitions. They do not use the forcing $q(t)$.

In this derivative-chain construction, only the final equation:

$$x'_n = y^{(n)}$$

uses the original differential equation. Since $q(t)$ appears in the formula for $y^{(n)}$, it enters only the final component of $\mathbf{f}(t)$.

Initial Data In State Order

Suppose the scalar problem provides:

$$y(t_0) = c_0, \quad y'(t_0) = c_1, \quad \dots, \quad y^{(n-1)}(t_0) = c_{n-1}.$$

The state definition gives:

$$\mathbf{x}(t_0) = \begin{bmatrix} y(t_0) \\ y'(t_0) \\ \vdots \\ y^{(n-1)}(t_0) \end{bmatrix}.$$

Substitute the prescribed values:

$$\boxed{\mathbf{x}(t_0) = \begin{bmatrix} c_0 \\ c_1 \\ \vdots \\ c_{n-1} \end{bmatrix}}.$$

The order of the entries must match the order chosen in $\mathbf{x}$.

Block 3 Summary

After normalizing the scalar equation, the companion matrix has:

- 1s immediately above the main diagonal
- zeros elsewhere in the first $n - 1$ rows
- negated normalized coefficients in the final row
- the normalized forcing in the final entry of $\mathbf{f}(t)$

The construction is a compressed record of the scalar derivative chain.

Variable Coefficients And A Third-Order Example

The Goal Of This Block

The goal is to show that the conversion works when coefficients depend on t , and to make every entry of a 3×3 system traceable.

The Initial-Value Problem

Convert:

$$u''' + 2t u'' - (t + 1)u' + 5u = e^{-t},$$

with:

$$u(1) = 2, \quad u'(1) = 0, \quad u''(1) = -1.$$

The leading coefficient is already 1, so no division is required.

Step 1: Isolate The Third Derivative

Start from:

$$u''' + 2t u'' - (t + 1)u' + 5u = e^{-t}.$$

Move every term except u''' to the right. Each moved term changes sign:

$$u''' = -5u + (t + 1)u' - 2t u'' + e^{-t}.$$

The coefficient of u' becomes positive because its original coefficient was $-(t + 1)$.

Step 2: Build The Three State Equations

Define:

$$x_1 = u, \quad x_2 = u', \quad x_3 = u''.$$

Differentiate $x_1 = u$ and use $x_2 = u'$:

$$x_1' = u' = x_2.$$

Differentiate $x_2 = u'$ and use $x_3 = u''$:

$$x_2' = u'' = x_3.$$

Differentiate $x_3 = u''$:

$$x_3' = u'''.$$

Substitute the isolated formula for u''' :

$$x'_3 = -5u + (t+1)u' - 2t u'' + e^{-t}.$$

Replace u, u', u'' by x_1, x_2, x_3 :

$$\boxed{x'_3 = -5x_1 + (t+1)x_2 - 2t x_3 + e^{-t}}.$$

Step 3: Assemble The Matrix System

Write the scalar equations with all state coefficients visible:

$$x'_1 = 0x_1 + 1x_2 + 0x_3,$$

$$x'_2 = 0x_1 + 0x_2 + 1x_3,$$

$$x'_3 = -5x_1 + (t+1)x_2 - 2t x_3 + e^{-t}.$$

Reading the coefficient of x_1, x_2, x_3 across each row gives:

$$\boxed{\mathbf{x}' = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -5 & t+1 & -2t \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 0 \\ e^{-t} \end{bmatrix}}.$$

Because two entries depend on t , this is a variable-coefficient matrix system.

Step 4: Assemble The Initial State

The state vector is:

$$\mathbf{x}(t) = \begin{bmatrix} u(t) \\ u'(t) \\ u''(t) \end{bmatrix}.$$

Evaluate at the prescribed initial time $t_0 = 1$:

$$\mathbf{x}(1) = \begin{bmatrix} u(1) \\ u'(1) \\ u''(1) \end{bmatrix}.$$

Substitute the three initial values:

$$\mathbf{x}(1) = \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix}.$$

The initial time remains 1; converting to a system does not shift it to zero.

A Leading Coefficient That Can Vanish

Consider:

$$t^2 y'' + 3ty' - 2y = \ln t.$$

To isolate y'' , divide by t^2 :

$$y'' = -\frac{3}{t}y' + \frac{2}{t^2}y + \frac{\ln t}{t^2}.$$

This normalization requires $t \neq 0$. The forcing $\ln t$ additionally requires $t > 0$ for real-valued solutions. Therefore the converted system is considered on an interval contained in:

$$t > 0.$$

The practical meaning:

dividing by the leading coefficient can reveal interval restrictions that must accompany the matrix system.

Block 4 Summary

Variable coefficients become entries of $A(t)$. A nonzero forcing becomes an entry of $f(t)$. Neither feature changes the derivative-chain rows.

Before dividing by a leading coefficient, identify where it is nonzero and retain that interval information in the converted problem.

Coupled Equations Of Different Orders

The Goal Of This Block

The goal is to create one state variable for every required derivative across several unknown functions and preserve all cross-coupling terms.

How Many State Variables Are Needed?

Suppose one equation is second order in $u(t)$ and another is first order in $v(t)$. The state must remember:

$$u, \quad u', \quad v.$$

Therefore three state variables are needed.

More generally, add the orders of the unknown functions. A third-order equation in u coupled to a second-order equation in v normally needs:

$$3 + 2 = 5$$

state variables.

A Mixed-Order Coupled Problem

Convert:

$$\begin{aligned}u'' + 2u' - v &= \sin t, \\v' + 3v - 4u &= t,\end{aligned}$$

with:

$$u(0) = 1, \quad u'(0) = 0, \quad v(0) = -2.$$

Use the state ordering:

$$x_1 = u, \quad x_2 = u', \quad x_3 = v.$$

Step 1: Isolate The Needed Derivatives

From the first equation:

$$u'' + 2u' - v = \sin t.$$

Move $2u'$ and $-v$ to the right:

$$u'' = -2u' + v + \sin t.$$

From the second equation:

$$v' + 3v - 4u = t.$$

In $v' + 3v - 4u = t$, move $3v$ and $-4u$ to the right:

$$v' = 4u - 3v + t.$$

Both isolated formulas are needed because they supply derivatives of state components.

Step 2: Translate Each Derivative Into State Variables

Differentiate $x_1 = u$:

$$x'_1 = u' = x_2.$$

Differentiate $x_2 = u'$ and use the isolated formula for u'' :

$$\begin{aligned} x'_2 &= u'' \\ &= -2u' + v + \sin t \\ &= -2x_2 + x_3 + \sin t. \end{aligned}$$

Differentiate $x_3 = v$ and use the isolated formula for v' :

$$\begin{aligned} x'_3 &= v' \\ &= 4u - 3v + t \\ &= 4x_1 - 3x_3 + t. \end{aligned}$$

Every original unknown has now been replaced by a state component.

Step 3: Read Off The Matrix And Forcing

Make all zero coefficients visible:

$$\begin{aligned}x'_1 &= 0x_1 + 1x_2 + 0x_3 + 0, \\x'_2 &= 0x_1 - 2x_2 + 1x_3 + \sin t, \\x'_3 &= 4x_1 + 0x_2 - 3x_3 + t.\end{aligned}$$

Reading across the rows gives:

$$\mathbf{x}' = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -2 & 1 \\ 4 & 0 & -3 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ \sin t \\ t \end{bmatrix}.$$

The cross-coupling term v creates the entry 1 in row 2, column 3. The cross-coupling term $4u$ creates the entry 4 in row 3, column 1.

Step 4: Convert The Initial Data

The selected state ordering is:

$$\mathbf{x}(t) = \begin{bmatrix} u(t) \\ u'(t) \\ v(t) \end{bmatrix}.$$

At $t = 0$:

$$\mathbf{x}(0) = \begin{bmatrix} u(0) \\ u'(0) \\ v(0) \end{bmatrix}.$$

Substitute the prescribed values:

$$\mathbf{x}(0) = \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}.$$

Two Second-Order Unknowns

For two second-order unknowns, a useful ordering is:

$$x_1 = u, \quad x_2 = u', \quad x_3 = v, \quad x_4 = v'.$$

Then:

$$x'_1 = x_2, \quad x'_3 = x_4.$$

The two original equations must be rearranged to provide formulas for:

$$x'_2 = u'', \quad x'_4 = v''.$$

The state order need not place all unprimed functions first. What matters is that every matrix row and every initial value follows the declared order consistently.

Block 5 Summary

For a coupled system:

1. count the derivative levels required for each unknown
2. declare a state ordering explicitly
3. isolate every highest derivative needed by that state
4. translate each equation into state components
5. preserve cross-coupling coefficients in the correct columns
6. place all non-state terms into the forcing vector

Equivalence, Reverse Conversion, And The Solution Bridge

The Goal Of This Block

The goal is to read a companion system back as a scalar equation and connect the converted system with the matrix exponential from Chapter 16.

Recover A Scalar Equation From A System

Suppose:

$$\mathbf{x}' = \begin{bmatrix} 0 & 1 \\ -6 & -5 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ e^t \end{bmatrix},$$

where:

$$\mathbf{x} = \begin{bmatrix} y \\ y' \end{bmatrix}.$$

Multiply the first row:

$$y' = 0y + 1y'.$$

This row confirms the state definition.

Multiply the second row. Since the second component derivative is y'' :

$$y'' = -6y - 5y' + e^t.$$

From $y'' = -6y - 5y' + e^t$, move the y and y' terms to the left:

$$\boxed{y'' + 5y' + 6y = e^t}.$$

The final row of a companion system contains the scalar equation in isolated highest-derivative form.

Not Every Matrix System Is In Companion Form

The system:

$$\mathbf{z}' = \begin{bmatrix} 2 & 1 \\ -3 & 4 \end{bmatrix} \mathbf{z}$$

is a valid first-order system, but its first row is:

$$z_1' = 2z_1 + z_2,$$

not $z_1' = z_2$. Therefore z_2 has not been declared to be z_1' .

It may still be possible to eliminate z_2 and obtain a second-order equation, but the direct companion reading rule does not apply until the state relationship is established.

The Homogeneous Constant-Coefficient Bridge

If conversion produces:

$$\mathbf{x}' = A\mathbf{x}, \quad \mathbf{x}(t_0) = \mathbf{c},$$

For this constant homogeneous system, Chapter 16 gives:

$$\mathbf{x}(t) = e^{A(t-t_0)} \mathbf{c}.$$

The first component of $\mathbf{x}(t)$ is the original unknown $y(t)$ when $x_1 = y$.

The conversion and the matrix exponential perform different jobs:

- conversion constructs the first-order system
- the matrix exponential solves a constant homogeneous system

The Forced Constant-Coefficient Bridge

For:

$$\mathbf{x}' = A\mathbf{x} + \mathbf{f}(t), \quad \mathbf{x}(t_0) = \mathbf{c},$$

with constant A , the solution can be written as:

$$\mathbf{x}(t) = e^{A(t-t_0)} \mathbf{c} + \int_{t_0}^t e^{A(t-s)} \mathbf{f}(s) ds.$$

The first term carries the initial state. The integral accumulates the effect of the forcing from t_0 to t .

Verify The Forced Formula

Let:

$$\mathbf{x}(t) = e^{A(t-t_0)} \mathbf{c} + \int_{t_0}^t e^{A(t-s)} \mathbf{f}(s) ds.$$

At $t = t_0$, the integral has equal limits and is zero:

$$\begin{aligned}\mathbf{x}(t_0) &= e^{A0}\mathbf{c} + 0 \\ &= I\mathbf{c} \\ &= \mathbf{c}.\end{aligned}$$

Now differentiate. The first term gives:

$$\frac{d}{dt} (e^{A(t-t_0)}\mathbf{c}) = Ae^{A(t-t_0)}\mathbf{c}.$$

For the integral, the upper-limit term is:

$$e^{A(t-t)}\mathbf{f}(t) = I\mathbf{f}(t) = \mathbf{f}(t).$$

Differentiating the integrand with respect to t gives:

$$\frac{\partial}{\partial t} (e^{A(t-s)}\mathbf{f}(s)) = Ae^{A(t-s)}\mathbf{f}(s).$$

Therefore:

$$\begin{aligned}\mathbf{x}'(t) &= Ae^{A(t-t_0)}\mathbf{c} + \mathbf{f}(t) + \int_{t_0}^t Ae^{A(t-s)}\mathbf{f}(s) ds \\ &= A \left[e^{A(t-t_0)}\mathbf{c} + \int_{t_0}^t e^{A(t-s)}\mathbf{f}(s) ds \right] + \mathbf{f}(t) \\ &= A\mathbf{x}(t) + \mathbf{f}(t).\end{aligned}$$

Both the differential equation and initial condition are satisfied.

Block 6 Summary

A companion system can be read backward because its derivative-chain rows identify each state component with a derivative of the first component.

For constant A , Chapter 16 supplies the solution operator. Conversion is therefore the bridge from a higher-order scalar equation to matrix methods.

A Reliable Workflow And Common Mistakes

The Goal Of This Block

The goal is to turn the conversion into a repeatable checklist and diagnose the errors that most often make a plausible-looking matrix incorrect.

A Reliable Conversion Workflow

Step 1: Count The Required State Components. An n th-order scalar equation needs n components. For coupled equations, add the derivative levels needed for every unknown.

Step 2: Declare The State Order. Write the full definition before constructing a matrix:

$$\mathbf{x} = [x_1, x_2, \dots, x_N]^T.$$

Step 3: Normalize Each Highest Derivative. Divide by its leading coefficient where that coefficient is nonzero. Record any interval restriction.

Step 4: Isolate Every Required Highest Derivative. Move all remaining terms carefully and show their sign changes.

Step 5: Write Scalar First-Order Equations. Do not jump directly to the matrix. First write one equation for each component derivative.

Step 6: Replace Original Variables By State Components. No y'' , u' , or other undeclared derivative should remain on the right side.

Step 7: Read Coefficients Row By Row. Use the same component order in every row. Write zeros explicitly when they help expose the structure.

Step 8: Separate The Forcing. Terms depending only on t , constants, or prescribed inputs belong in $\mathbf{f}(t)$, not in $\mathbf{A}(t)$.

Step 9: Translate Initial Data In State Order. Keep the original initial time and place each value in its matching component.

Step 10: Expand The Matrix Product To Verify. Recover the scalar equations row by row and compare them with the isolated derivative formulas.

Common Mistake: Reading Coefficients Before Isolating

From:

$$y'' + 4y' - 7y = h(t),$$

the final state equation is not obtained by copying 4 and -7 directly.

First isolate y'' :

$$y'' = -4y' + 7y + h(t).$$

With $x_1 = y$ and $x_2 = y'$, write in state order:

$$x_2' = 7x_1 - 4x_2 + h(t).$$

Because $x_2' = 7x_1 - 4x_2 + h(t)$, the final matrix row is:

$$\begin{bmatrix} 7 & -4 \end{bmatrix}.$$

Common Mistake: Treating Forcing As A Matrix Coefficient

For:

$$x_2' = 3x_1 - x_2 + t^2,$$

the term t^2 multiplies no state variable. It therefore belongs in the forcing vector:

$$\begin{bmatrix} x_1' \\ x_2' \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ t^2 \end{bmatrix}.$$

Common Mistake: Losing The State Order

Suppose:

$$\mathbf{x} = \begin{bmatrix} u \\ u' \\ v \end{bmatrix}.$$

The initial vector must use that same order:

$$\mathbf{x}(t_0) = \begin{bmatrix} u(t_0) \\ u'(t_0) \\ v(t_0) \end{bmatrix}.$$

Writing $[u(t_0), v(t_0), u'(t_0)]^T$ silently assigns values to the wrong components.

Common Mistake: Changing The Initial Time

If the scalar problem gives data at $t_0 = 3$, the system condition is:

$$\mathbf{x}(3) = \mathbf{c},$$

not $\mathbf{x}(0) = \mathbf{c}$.

For a constant homogeneous system, the matching solution formula is:

$$\mathbf{x}(t) = e^{A(t-3)}\mathbf{c}.$$

The shift $t - 3$ is forced by the original initial time.

Common Mistake: Using The Wrong Number Of Components

A fourth-order scalar equation needs four first-order state equations. Using only y, y', y'' leaves no component whose derivative is $y^{(4)}$.

The correct chain is:

$$x_1 = y, \quad x_2 = y', \quad x_3 = y'', \quad x_4 = y''''.$$

Then $x_4' = y^{(4)}$ can be supplied by the original equation.

Block 7 Summary

The most reliable habit is:

write and verify the scalar first-order equations before compressing them into a matrix.

Most matrix errors are really earlier errors in normalization, signs, state ordering, or forcing separation.

Original Practice Set

How To Use These Problems

For each conversion:

1. declare the state variables
2. isolate the required highest derivative
3. write all scalar first-order equations
4. assemble $A(t)$ and $f(t)$
5. translate initial data when provided
6. expand at least one matrix row as a check

The twelve problems below use equations and data created specifically for this chapter.

Problems 1 To 4: Single Scalar Equations

Problem 1: A Forced Second-Order IVP. Convert:

$$y'' + 7y' - 2y = t^2, \quad y(0) = 3, \quad y'(0) = -1.$$

Problem 2: Normalize Before Reading Variable Coefficients. On an interval containing $t = 1$, convert:

$$3y'' - 6ty' + 9y = \cos t, \quad y(1) = 0, \quad y'(1) = 2.$$

Problem 3: A Third-Order Equation. Convert:

$$z''' - 4z'' + z' + 6z = e^t.$$

Problem 4: A Fourth-Order Equation With Missing Derivatives. Convert:

$$w^{(4)} + 2w'' - w = t.$$

Problems 5 To 8: Domains, Reversal, And Diagnosis

Problem 5: Record The Domain Restriction. Convert on an appropriate real interval:

$$t y'' + (1 - t)y' + 4y = \sqrt{t}.$$

Problem 6: Recover The Scalar Equation. Given:

$$\mathbf{x}' = \begin{bmatrix} 0 & 1 \\ 8 & -6 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 3 \sin t \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} y \\ y' \end{bmatrix},$$

recover the scalar differential equation.

Problem 7: Diagnose A Sign Error. A learner converts:

$$y'' - 5y' + 2y = 0$$

using $x_1 = y, x_2 = y'$ and proposes the final row $[-2, -5]$. Identify the error and give the correct matrix.

Problem 8: Put An Existing First-Order System In Matrix Form. Write:

$$u' = 2u - 3v + e^{-t},$$

$$v' = 5u + v - 4$$

as $\mathbf{x}' = A\mathbf{x} + \mathbf{f}(t)$ using $\mathbf{x} = [u, v]^T$.

Problems 9 To 12: Coupling And The Solution Bridge**Problem 9: One Second-Order And One First-Order Unknown.** Convert:

$$\begin{aligned}p'' - p' + 2q &= t, \\ q' + q - 3p &= e^t,\end{aligned}$$

using $\mathbf{x} = [p, p', q]^\top$.**Problem 10: Two Second-Order Unknowns.** Convert:

$$\begin{aligned}u'' + 2u - v' &= 0, \\ v'' - u' + 3v &= \cos t,\end{aligned}$$

using $\mathbf{x} = [u, u', v, v']^\top$.**Problem 11: Translate Mixed Initial Data.** For the state order:

$$\mathbf{x} = [r, r', r'', s, s']^\top,$$

write the initial vector when:

$$r(2) = -1, \quad r'(2) = 4, \quad r''(2) = 0, \quad s(2) = 3, \quad s'(2) = -2.$$

Problem 12: Use The Matrix-Exponential Bridge. The equation:

$$y'' - y = 0, \quad y(2) = 3, \quad y'(2) = 1$$

is converted using $\mathbf{x} = [y, y']^\top$. Write the matrix IVP and its solution in matrix-exponential form. It is not necessary to calculate the entries of the exponential.

Worked Answers: Problems 1 To 4

Answer 1. Start from:

$$y'' + 7y' - 2y = t^2.$$

Isolate y'' :

$$y'' = -7y' + 2y + t^2.$$

Use:

$$x_1 = y, \quad x_2 = y'.$$

Differentiate the state definitions and substitute the isolated derivative:

$$\begin{aligned} x_1' &= x_2, \\ x_2' &= 2x_1 - 7x_2 + t^2. \end{aligned}$$

Therefore:

$$\mathbf{x}' = \begin{bmatrix} 0 & 1 \\ 2 & -7 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ t^2 \end{bmatrix}.$$

The state order is $[y, y']^T$, so:

$$\mathbf{x}(0) = \begin{bmatrix} y(0) \\ y'(0) \end{bmatrix} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}.$$

Answer 2. Start from:

$$3y'' - 6t y' + 9y = \cos t.$$

Move the lower-derivative terms to the right:

$$3y'' = 6t y' - 9y + \cos t.$$

Divide every term by 3:

$$y'' = -3y + 2t y' + \frac{1}{3} \cos t.$$

With $x_1 = y$ and $x_2 = y'$:

$$\begin{aligned} x_1' &= x_2, \\ x_2' &= -3x_1 + 2t x_2 + \frac{1}{3} \cos t. \end{aligned}$$

Thus:

$$\mathbf{x}' = \begin{bmatrix} 0 & 1 \\ -3 & 2t \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ \frac{1}{3} \cos t \end{bmatrix}.$$

At the original initial time:

$$\mathbf{x}(1) = \begin{bmatrix} y(1) \\ y'(1) \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}.$$

Answer 3. Isolate the third derivative in:

$$z''' - 4z'' + z' + 6z = e^t.$$

Moving the remaining terms gives:

$$z''' = -6z - z' + 4z'' + e^t.$$

Define:

$$x_1 = z, \quad x_2 = z', \quad x_3 = z''.$$

The scalar state equations are:

$$x_1' = x_2,$$

$$x_2' = x_3,$$

$$x_3' = -6x_1 - x_2 + 4x_3 + e^t.$$

Reading the coefficients in the order x_1, x_2, x_3 gives:

$$\mathbf{x}' = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -1 & 4 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 0 \\ e^t \end{bmatrix}.$$

Answer 4. Start from:

$$w^{(4)} + 2w'' - w = t.$$

For $w^{(4)} + 2w'' - w = t$, isolate the fourth derivative:

$$w^{(4)} = w - 2w'' + t.$$

Define four components:

$$x_1 = w, \quad x_2 = w', \quad x_3 = w'', \quad x_4 = w''''.$$

Differentiate each definition and use the isolated equation in the final row:

$$x_1' = x_2,$$

$$x_2' = x_3,$$

$$x_3' = x_4,$$

$$x_4' = x_1 - 2x_3 + t.$$

Make the missing coefficients visible:

$$x_4' = 1x_1 + 0x_2 - 2x_3 + 0x_4 + t.$$

Therefore:

$$\mathbf{x}' = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & -2 & 0 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ t \end{bmatrix}.$$

Worked Answers: Problems 5 To 8

Answer 5. The equation is:

$$t y'' + (1 - t)y' + 4y = \sqrt{t}.$$

For real $\sqrt{t}$, require $t \geq 0$. Isolating y'' also requires $t \neq 0$. Together these restrictions give:

$$t > 0.$$

Move the lower-derivative terms:

$$t y'' = -(1 - t)y' - 4y + \sqrt{t}.$$

Divide every term by t :

$$y'' = -\frac{4}{t}y - \frac{1-t}{t}y' + \frac{\sqrt{t}}{t}.$$

With $x_1 = y$ and $x_2 = y'$:

$$\begin{aligned} x_1' &= x_2, \\ x_2' &= -\frac{4}{t}x_1 - \frac{1-t}{t}x_2 + \frac{\sqrt{t}}{t}. \end{aligned}$$

Hence:

$$\mathbf{x}' = \begin{bmatrix} 0 & 1 \\ -\frac{4}{t} & -\frac{1-t}{t} \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ \frac{\sqrt{t}}{t} \end{bmatrix}, \quad t > 0.$$

Answer 6. Use the declared state:

$$\mathbf{x} = \begin{bmatrix} y \\ y' \end{bmatrix}.$$

The second row of the matrix system is:

$$y'' = 8y - 6y' + 3 \sin t.$$

Move the state-dependent terms to the left:

$$y'' + 6y' - 8y = 3 \sin t.$$

The first row gives $y' = y'$, confirming the derivative-chain ordering.

Answer 7. Start from:

$$y'' - 5y' + 2y = 0.$$

Isolate y'' . The lower-order terms move right and change signs:

$$y'' = 5y' - 2y.$$

Write the terms in state order y, y' :

$$y'' = -2y + 5y'.$$

Therefore the final row is $[-2, 5]$, not $[-2, -5]$. The correct matrix is:

$$\mathbf{x}' = \begin{bmatrix} 0 & 1 \\ -2 & 5 \end{bmatrix} \mathbf{x}.$$

The proposed conversion failed to change the sign of $-5y'$.

Answer 8. Use:

$$\mathbf{x} = \begin{bmatrix} u \\ v \end{bmatrix}, \quad \mathbf{x}' = \begin{bmatrix} u' \\ v' \end{bmatrix}.$$

Keep state terms and forcing terms separate:

$$u' = 2u - 3v + e^{-t},$$

$$v' = 5u + 1v - 4.$$

Therefore:

$$\mathbf{x}' = \begin{bmatrix} 2 & -3 \\ 5 & 1 \end{bmatrix} \mathbf{x} + \begin{bmatrix} e^{-t} \\ -4 \end{bmatrix}.$$

Worked Answers: Problems 9 To 12

Answer 9. Use the prescribed state order:

$$x_1 = p, \quad x_2 = p', \quad x_3 = q.$$

Isolate the required derivatives. From:

$$p'' - p' + 2q = t$$

obtain:

$$p'' = p' - 2q + t.$$

From:

$$q' + q - 3p = e^t$$

obtain:

$$q' = 3p - q + e^t.$$

Translate the isolated formulas:

$$x_1' = x_2,$$

$$x_2' = x_2 - 2x_3 + t,$$

$$x_3' = 3x_1 - x_3 + e^t.$$

Make every state coefficient visible:

$$\begin{aligned}x_1' &= 0x_1 + 1x_2 + 0x_3, \\x_2' &= 0x_1 + 1x_2 - 2x_3 + t, \\x_3' &= 3x_1 + 0x_2 - 1x_3 + e^t.\end{aligned}$$

Therefore:

$$\mathbf{x}' = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & -2 \\ 3 & 0 & -1 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ t \\ e^t \end{bmatrix}.$$

Answer 10. Use:

$$x_1 = u, \quad x_2 = u', \quad x_3 = v, \quad x_4 = v'.$$

From $u'' + 2u - v' = 0$, isolate:

$$u'' = -2u + v'.$$

From $v'' - u' + 3v = \cos t$, isolate:

$$v'' = u' - 3v + \cos t.$$

Translate all four component derivatives:

$$\begin{aligned}x_1' &= x_2, \\x_2' &= -2x_1 + x_4, \\x_3' &= x_4, \\x_4' &= x_2 - 3x_3 + \cos t.\end{aligned}$$

Reading coefficients in the order x_1, x_2, x_3, x_4 gives:

$$\mathbf{x}' = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -2 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & -3 & 0 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ \cos t \end{bmatrix}.$$

For example, row 2 expands to:

$$x_2' = -2x_1 + x_4,$$

which restores to $u'' = -2u + v'$.

Answer 11. The declared state order is:

$$\mathbf{x} = [r, r', r'', s, s']^\top.$$

Evaluate the same ordering at $t = 2$:

$$\mathbf{x}(2) = \begin{bmatrix} r(2) \\ r'(2) \\ r''(2) \\ s(2) \\ s'(2) \end{bmatrix}.$$

Substitute each given value in its matching position:

$$\mathbf{x}(2) = \begin{bmatrix} -1 \\ 4 \\ 0 \\ 3 \\ -2 \end{bmatrix}.$$

Answer 12. Start from:

$$y'' - y = 0.$$

Isolate the second derivative:

$$y'' = y.$$

With $x_1 = y$ and $x_2 = y'$:

$$x'_1 = x_2,$$

$$x'_2 = x_1.$$

Therefore the matrix IVP is:

$$\mathbf{x}' = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \mathbf{x}, \quad \mathbf{x}(2) = \begin{bmatrix} 3 \\ 1 \end{bmatrix}.$$

Let:

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \mathbf{c} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}.$$

For a constant homogeneous system with $t_0 = 2$, Chapter 16 gives:

$$\mathbf{x}(t) = e^{A(t-2)} \mathbf{c}.$$

Substitute A and $\mathbf{c}$:

$$\mathbf{x}(t) = \exp\left(\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} (t-2)\right) \begin{bmatrix} 3 \\ 1 \end{bmatrix}.$$

The first component of $\mathbf{x}(t)$ is the original function $y(t)$.

Chapter 17 Final Summary

An n th-order scalar equation becomes an n -component first-order system by using:

$$\boxed{x_1 = y, \quad x_2 = y', \quad \dots, \quad x_n = y^{(n-1)}}.$$

For the normalized equation:

$$y^{(n)} + p_{n-1}(t)y^{(n-1)} + \dots + p_1(t)y' + p_0(t)y = q(t),$$

the scalar first-order equations are:

$$\begin{aligned} x_1' &= x_2, \\ x_2' &= x_3, \\ &\vdots \\ x_{n-1}' &= x_n, \\ x_n' &= -p_0x_1 - p_1x_2 - \dots - p_{n-1}x_n + q. \end{aligned}$$

The companion matrix is:

$$A(t) = \begin{bmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \\ -p_0 & -p_1 & -p_2 & \dots & -p_{n-1} \end{bmatrix}.$$

Initial values are collected in state order:

$$\boxed{\mathbf{x}(t_0) = \begin{bmatrix} y(t_0) \\ y'(t_0) \\ \vdots \\ y^{(n-1)}(t_0) \end{bmatrix}}.$$

For coupled equations, create enough components to represent every unknown and every derivative level required by its order. Cross-coupling terms become off-diagonal matrix entries.

For constant A , the converted homogeneous system connects directly to:

$$\mathbf{x}' = A\mathbf{x}, \quad \mathbf{x}(t_0) = \mathbf{c} \quad \Rightarrow \quad \boxed{\mathbf{x}(t) = e^{A(t-t_0)}\mathbf{c}}.$$

The chapter's central lesson is:

declare the state order, isolate each highest derivative, write the scalar first-order equations, and only then read off the matrix row by row.